

BIO Framework: The Council's Definitive Critical Review

Síntese da avaliação coletiva de um “conselho” de múltiplas IAs/LLMs sobre o BIO v0.1. Colado pelo usuário na conversa de origem. Documento crítico-chave do projeto.

Chairman's Preface

After reviewing all responses and their peer rankings, a clear consensus emerges. The highest-ranked responses share four characteristics: rigorous persona-specific critique, identification of concrete prior art, systematic treatment of epistemological flaws, and brutal honesty about social/governance failures.

One meta-finding deserves emphasis: the hardest problems in BIO are not technical. They are social, epistemological, and economic. The technical challenges are real but solvable. The social challenges may not be.

QUESTION 1: Is BIO Genuinely Novel?

Verdict: Marginally novel as an integration. Not novel as a paradigm.

Prior art the manifesto does not acknowledge: - **Nanopublications** (since 2010): structured claims with provenance — directly analogous to BIO's core primitive - **Wikidata**: versioned, machine-readable statements with references and confidence ranks - **Cochrane Living Systematic Reviews** (since 2017): continuously updated evidence synthesis - **scite.ai**: already ships supporting/contradicting/mentioning citation classification - **GRADE**: structured evidence certainty assessment - **Open Targets, Monarch Initiative, SemMedDB**: biomedical knowledge graphs at scale - **ClinGen, CIViC, PharmGKB**: expert-curated, claim-level biomedical knowledge with governance

The failure to cite these is the manifesto's most serious credibility problem. What BIO might contribute — a generalized, disease-agnostic, claim-level infrastructure integrating these — is **infrastructural novelty, not conceptual novelty**.

QUESTION 2: The Strongest Arguments Against BIO

1. **BIO conflates consensus with truth (epistemological).** The history of science is a chronicle of consensus being wrong: continental drift, H. pylori, Semmelweis. A system that displays consensus as “confidence” penalizes correct heterodox views.
2. **Claims are not atomic, context-independent objects.** The meaning of a claim depends on diagnostic criteria, measurement, reference population, statistical model, study design. BIO will either create thousands of near-duplicate claims or collapse them misleadingly.

3. **BIO will suppress paradigm shifts (historical).** A confidence-score knowledge graph is an engine for normal science, structurally hostile to revolutionary science. Old claims don't get lower scores — they become incommensurable.
4. **The governance problem will kill BIO before the technology does.** Two experts disagree (one does liposuction, one conservative management). Who resolves? By what authority? Wikipedia took 20 years and norms are still contested.
5. **The incentive structure is fatally misaligned.** Scientists are rewarded for publications, not curating claims. The open-source analogy breaks: developers contribute because they use the software; scientists gain near-zero private benefit.
6. **The Git analogy is intellectually misleading.** There is no make test for a scientific claim. Merging “vascular origin” vs “metabolic origin” is a scientific judgment no algorithm resolves.

QUESTION 3: Assumptions Likely to Be Wrong

- Claims can be cleanly extracted by AI (biomedical NLP F1 ~0.6–0.7 on stance detection)
- Experts will contribute continuously (every collaborative platform faces retention problems)
- Consensus reflects epistemic validity (reflects social dynamics, funding, inertia)
- Version control maps onto scientific knowledge (knowledge is holistic, theoretically embedded)
- Niche field → generalization (lipedema's small, contested community is a hard case)
- Science is cumulative (paradigm shifts are discontinuous, incommensurable)
- A single confidence score represents evidence quality (it is irreducibly multidimensional)
- Disease-agnostic design achievable early (premature generalization → wrong for every domain)

QUESTION 4: Practical Obstacles to Adoption

Governance (no authority/dispute/COI model); Incentives (no career credit); Legal liability (confidence scores → product-liability exposure); Cold-start problem (no value until comprehensive, not comprehensive until funded); Interoperability (PubMed, MeSH, SNOMED CT, UMLS, ORCID, DOI, ClinicalTrials.gov, EHR); AI accuracy (2–5% error at scale = hundreds of thousands of false claims); Ontology maintenance (concepts change continuously).

QUESTION 5: Missing Incentives

Researchers need citable DOI per contribution, ORCID linkage, recognition by funders (NIH/ERC/Wellcome) and in tenure. Journals need a model where they participate (mandatory claim deposition) rather than compete. Institutions need funded maintenance roles (like Cochrane). **The fundamental missing element: BIO must be useful to contributors before it is useful to the community.**

QUESTION 6: How to Simplify While Preserving the Core Innovation

The core innovation worth preserving: **claim-level provenance with updateable evidence status.**

Minimal defensible version: - One disease (lipedema), 25–50 high-priority claims, manually selected - Each claim: canonical text, type, population/context, linked evidence with study design + risk-of-bias tags, modification history, curator attribution - No AI extraction (human curation until AI validated) - No novel confidence metrics (use GRADE) - No knowledge graph (simple hierarchical organization) - No real-time updates (scheduled cycles) - API on nanopublication structure

The test: Does this produce better knowledge outputs than a well-conducted Cochrane living review at comparable cost?

QUESTION 7: Essential vs. Unnecessary Complexity

Essential: claim-level persistent IDs; evidence provenance (study-level); version history with curator attribution; explicit uncertainty; human editorial authority; COI disclosure; interoperability (DOI, ORCID, MeSH/SNOMED CT). **Useful but not at launch:** AI surveillance; graph visualization; geographic opinion tracking. **Dangerous until validated:** unified confidence scores; consensus as primary validity signal; AI extraction without validated accuracy. **Probably unnecessary:** disease-agnostic architecture at v0.1; real-time updating everywhere; “Git for science” branding.

QUESTION 8: What Top-Journal Reviewers Would Say

“This is a manifesto, not a research article.” / “The novelty claim is unsupported.” / “The technical proposal does not exist.” / “The governance is entirely unspecified.” / “The AI component is asserted, not validated.” / “Lipedema as proof-of-concept is self-defeating (a hard case).” / “There is no comparison against the status quo.” / “The metrics are operationally undefined.”

QUESTION 9: What Would Demonstrate That BIO Works?

Minimum: working prototype, 50–100 lipedema claims; AI extraction validated vs human curators (published sensitivity/specificity); 12 months operation with documented engagement; demonstrated update-speed advantage. Investment-scale: confidence scores shown to be calibrated; BIO-identified gaps led to funded research; independent adoption; documented case of faster knowledge-change detection; comparison study on clinical decision accuracy.

Falsifiable core question: Does claim-centric knowledge organization produce better epistemic outcomes than publication-centric organization, at acceptable cost?

QUESTION 10: Greatest Long-Term Impact If Successful

Changing the unit of scientific accountability from the paper to the claim. Specific claims would carry persistent accountability: who asserted it, what evidence supported it over time, when contradictory evidence emerged, who updated the record. Enables real-time propagation of retractions, transparent evidence gaps, reduced evidence-to-practice lag, machine-readable infrastructure for AI clinical tools. Comparable ambition: PubMed. Probability low; magnitude high; expected value uncertain.

Final Council Assessment

The idea is worth investigating. The proposal is not yet ready to execute.

Five actions, in priority order: 1. Conduct a serious literature review; reposition vs nanopublications, living reviews, Wikidata, GRADE. Acknowledge what exists; articulate what BIO adds. 2. Design governance before technology. 3. Build the minimum viable prototype: one domain, 50 claims, 20 experts, 12 months, human curation only. Measure and publish everything — including failures. 4. Solve the credit problem (citable, ORCID-linked, recognized in academic evaluation). 5. Commission an AI extraction validation study before deploying AI at any scale.

Build the simplest possible thing that could work. Test it rigorously. Publish the results — including when they are negative. That is Living Science. The manifesto is not.